

Cover Letter

Dear Editors,

We submit our manuscript, "Module-level agreement is not replication: a reporting and reproducibility gate for public human intervertebral disc transcriptomes," for consideration as a Special Communication.

The manuscript addresses a practical problem in public-data biomedical informatics: module-level significance and direction agreement are often described as replication even when the same genes do not move consistently across cohorts. We performed a matched audit of public human intervertebral disc transcriptome cohorts and operationalized the distinction between exploratory, internally stable, contextually concordant, candidate-replicated and FDR-controlled claims.

The contribution is not a new statistical test or algorithm. It is an integrated methodological audit with a reusable claim ladder and a sensitivity battery spanning donor dependence, probe selection, gene identity, platform scale and FDR-family structure. The results show that module-level significance can coexist with gene-level concordance that does not exceed chance, and that interpretation depends strongly on these design and analysis choices.

The manuscript fits JBI's emphasis on generalizable methodological lessons for developers and users of biomedical informatics methods. The work is original, is not under consideration elsewhere, and all authors have approved the submission.

Sincerely,

Zhongyuan Liu